

Homework 5

Deadline: October 4

Write $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$; subsequential limits may be infinite.

Problem 1.

Define (s_n) by

$$s_{3k-2} = -2 + \frac{1}{k}, \quad s_{3k-1} = 1 - \frac{1}{k}, \quad s_{3k} = k \quad (k \in \mathbb{N}).$$

Find all subsequential limits in $\overline{\mathbb{R}}$, and determine $\liminf s_n$ and $\limsup s_n$. State whether (s_n) has a limit in $\overline{\mathbb{R}}$.

Problem 2.

Let (a_n) and (b_n) be bounded sequences, and let $r_n \geq 0$ with $r_n \rightarrow 0$. Suppose $a_n \leq b_n + r_n$ for all sufficiently large n . Prove that

$$\liminf a_n \leq \liminf b_n, \quad \limsup a_n \leq \limsup b_n.$$

Work directly with infima and suprema of sufficiently late tails; do not assume that (a_n) or (b_n) has an ordinary limit.

Problem 3.

Let (p_n) and (q_n) repeat the following blocks of four:

$$(p_n) = (1, -1, 3, 0, 1, -1, 3, 0, \dots), \\ (q_n) = (0, 4, -2, 2, 0, 4, -2, 2, \dots).$$

Set $u_n = p_n + 1/n$ and $v_n = q_n - 2/n$. Compute

$$(a) \liminf u_n + \liminf v_n, \quad (b) \liminf(u_n + v_n), \quad (c) \liminf u_n + \limsup v_n, \\ (d) \limsup(u_n + v_n), \quad (e) \limsup u_n + \limsup v_n.$$

There is no need to write a proof.

Problem 4.

Determine whether each series converges or diverges. In each case, name a valid criterion and supply the inequality or limit that makes it apply. There is no need to find an exact sum.

$$(a) \sum_{n=1}^{\infty} \frac{n+1}{n^2+3n+1}, \quad (b) \sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1},$$

Problem 5.

Let (b_n) be a real sequence with $0 < c \leq |b_n| \leq C$ for all sufficiently large n , where c, C are fixed constants. Prove that $\sum_{n=1}^{\infty} a_n$ converges absolutely if and only if $\sum_{n=1}^{\infty} a_n b_n$ does, for every real sequence (a_n) .

Show that the positive lower bound is essential: give bounded (b_n) with $b_n \neq 0$ for every n and a real sequence (a_n) such that one of these series converges absolutely and the other does not. Justify both claims.